Probability Theory - chapter 01

* When we define two random variables y_1 and y_2 over the same sample space, we can represent the "bivariate probability function" as;

$$\boxed{p(y_1, y_2) = P(Y_1 = y_1, Y_2 = y_2) \; ; -\infty < y_1 < \infty , -\infty < y_2 < \infty}$$

This is also called as "joint probability function".
(Note: This y_1 and y_2 are discrete random variables.)

* This joint probability function is sometimes called as "joint probability mass function".

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* For any random variables y_1 and y_2, we can denote the "joint (bivariate) distribution function" as;

$$\boxed{F(y_1, y_2) = P(Y_1 \leq y_1 , Y_2 \leq y_2) \; ; -\infty < y_1 < \infty , -\infty < y_2 < \infty}$$

* For discrete random variables y_1 and y_2,

$$F(y_1, y_2) = \sum_{t_1 \leq y_1} \sum_{t_2 \leq y_2} p(t_1, t_2)$$

* For continuous random variables y_1 and y_2,

$$F(y_1, y_2) = \int_{-\infty}^{y_1} \int_{-\infty}^{y_2} f(t_1, t_2) \, dt_2 \, dt_1$$

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* For continuous random variables y_1 and y_2, "joint (bivariate) probability density function" is given by;

$f(y_1, y_2)$; $-\infty < y_1 < \infty$, $-\infty < y_2 < \infty$

Then ;

$$\boxed{F(y_1, y_2) = \int_{-\infty}^{y_1} \int_{-\infty}^{y_2} f(y_1, y_2)\, dy_2\, dy_1}$$

Marginal and Conditional probability distributions

* when y_1 and y_2 are discrete random variables with probability function $p(y_1, y_2)$; marginal probability functions ;

$$\boxed{p_1(y_1) = \sum_{\text{all } y_2} p(y_1, y_2)} \quad \text{and} \quad \boxed{p_2(y_2) = \sum_{\text{all } y_1} p(y_1, y_2)}$$

* when y_1 and y_2 are continuous random variables with joint density function $f(y_1, y_2)$; marginal density functions ;

$$\boxed{f_1(y_1) = \int_{-\infty}^{\infty} f(y_1, y_2)\, dy_2} \quad \text{and} \quad \boxed{f_2(y_2) = \int_{-\infty}^{\infty} f(y_1, y_2)\, dy_1}$$

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* Conditional probability function $\Rightarrow P(A \cap B) = P(A)\, P(B|A)$

$$\boxed{P(B|A) = \frac{P(A \cap B)}{P(A)}}$$

* Conditional discrete probability function y_1 given y_2 is ;

$$p(y_1|y_2) = p(Y_1 = y_1 \mid Y_2 = y_2) = \frac{p(Y_1 = y_1, Y_2 = y_2)}{p_2(y_2)} \quad ; p_2(y_2) > 0$$

* Conditional distribution function ;

$$F(y_1|y_2) = P(Y_1 \leq y_1 \mid Y_2 = y_2)$$

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* **Conditional density of y_1 given y_2 is;**

$$\boxed{f(y_1|y_2) = \frac{f(y_1, y_2)}{f_2(y_2)}} \; ; \; f_2(y_2) > 0$$

* **Conditional distribution function;**

$$\boxed{F(y_1|y_2) = \int_{-\infty}^{y_1} \frac{f(y_1, y_2)}{f_2(y_2)} \, dy_1 = f(y_1|y_2)}$$

Independent random variables, Expected value

* If A and B are independent;

$$P(A \cap B) = P(A) \times P(B)$$

* If $a < b$ and $c < d$ are in the type of $(a < Y_1 \leq b) \cap (c < Y_2 \leq d)$

$$P(a < Y_1 \leq b, c < Y_2 \leq d) = P(a < Y_1 \leq b) \times P(c < Y_2 \leq d)$$

Note: if Y_1 and Y_2 are independent; the joint probability is written as the product of marginal probabilities.

* So; Y_1 have distribution function $F_1(Y_1)$ and Y_2 have distribution function $F_2(Y_2)$. Also, Y_1 and Y_2 have joint distribution function $F(y_1, y_2)$.

Then, Y_1 and Y_2 are independent iff;

$$\boxed{F(y_1, y_2) = F_1(y_1) F_2(y_2)} \; ; \; y_1, y_2 \in \mathbb{R}$$

* when y_1 and y_2 are discrete random variables with joint probability function $p(y_1, y_2)$ and marginal probability functions $P_1(y_1)$ and $P_2(y_2)$. y_1 and y_2 are independent iff;

$$\boxed{P(y_1, y_2) = P_1(y_1) P_2(y_2)} \; ; \; y_1, y_2 \in \mathbb{R}$$

★ when y_1 and y_2 are continuous random variables with joint density function $f(y_1, y_2)$ and marginal density functions $f_1(y_1)$ and $f_2(y_2)$. y_1 and y_2 are independent iff;

$$\boxed{f(y_1, y_2) = f_1(y_1) f_2(y_2)} \; ; \; y_1, y_2 \in \mathbb{R}$$

★ $f(y_1, y_2)$ is the joint density function of y_1 and y_2.
$f(y_1, y_2) > 0$ iff $a \leq y_1 \leq b$, $c \leq y_2 \leq d$; otherwise $f(y_1, y_2) = 0$
y_1 and y_2 are independent iff,

$$f(y_1, y_2) = g(y_1) h(y_2)$$

$g(y_1) \rightarrow$ non-negative function of y_1 alone

$h(y_2) \rightarrow$ non-negative function of y_2 alone.

★ Expected value of a function is given by $E(x)$ and variance is given by $V(x)$:

$$\boxed{E(y_1) = \int_{-\infty}^{\infty} y_1 f_1(y_1) \, dy_1}$$

$$\boxed{V(y_1) = E(y_1^2) - [E(y_1)]^2}$$

$$\boxed{E(y_1) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y_1 f(y_1, y_2) \, dy_2 \, dy_1}$$

★ Special theorems;

① $\boxed{E(c) = c}$

② $\boxed{E[c\, g(y_1, y_2)] = c\, E[g(y_1, y_2)]}$

③ $\boxed{E[g_1(y_1, y_2) + g_2(y_1, y_2) + \dots + g_k(y_1, y_2)] = E[g_1(y_1, y_2)] + E[g_2(y_1, y_2)] + \dots + E[g_k(y_1, y_2)]}$

from ③;

$\therefore$ $E(y_1 - y_2) = E(y_1) - E(y_2)$

★ y_1, y_2 are independent ;

$$\boxed{E(y_1, y_2) = E(y_1)\,E(y_2)}$$

$$\boxed{E[g(y_1)h(y_2)] = E[g(y_1)]\,E[h(y_2)]}$$

Covariance

★ when y_1 and y_2 are random variables and μ_1 and μ_2 are means ;

$$\boxed{Cov(y_1, y_2) = E[(y_1-\mu_1)(y_2-\mu_2)]}$$

$$\boxed{\rho = \frac{Cov(y_1, y_2)}{\sigma_1 \sigma_2}}$$

Correlation coefficient

$$\boxed{Cov(y_1, y_2) = E(y_1 y_2) - \mu_1 \mu_2}$$

★ If y_1 and y_2 are independent ;

$Cov(y_1, y_2) = 0$ $\Rightarrow$ uncorrelated